Korobochka (コロボ) @cirnosad

Apr 20, 2025 · 1 tweets · cirnosad/status/1914008685577478653

Why are the primary colors Cyan, Magenta and Yellow and not Red, Yellow and Blue? [Well, the got yellow right at least.]

Take a look at this triple spectrum figure...

This corresponds to the three cones in our eyes that sense blue, green and red wavelengths from left to right. These are sometimes called the S-M-L cones (short-medium-long). They do not merely sense one wavelength though, they are sensitive to a range of wavelengths *each*. In turn, your vision (processing in the brain) interprets the mixture your sights sense, and creates the illusion of a continuous color spectrum. This is for given lighting conditions and other complex temporal and spatial sight/visual interaction (that often form optical illusions).

When you're looking at a smartphone, it's actually being sneaky and only outputting red, green and blue (RGB) light in different mixtures to make your brain see a certain color. If you get really close to a low-res screen, you'll be able to see the red green and blue dots. This is called "additive" color mixing.

How do we create colors on paper by mixing pigments/dyes that have fixed absorption/reflection profiles. We have to absorb the "humps" in our visual spectrum. A yellow pigment will kill the light reflected into the blue cone. A magnet pigment will kill the light reflected into the green cone. A cyan pigment will kill the light reflected into the red cone.

Mix all three and you get... black! That's because none of the colors get reflected back into your eyes. :) Leave the pigments out on a white sheet, you get white (obviously, lol). Now we can find whatever mixture we want and create any color we want.

Why couldn't they teach this to us in primary school? Because the education system is meant to make us stupid and helpless. I will never forgive them for robbing me of this intuition since I love art so much and I am completely useless at colouring and shading today.

Korobochka (コロボ) 🇦🇺✝️
@cirnosad · Follow

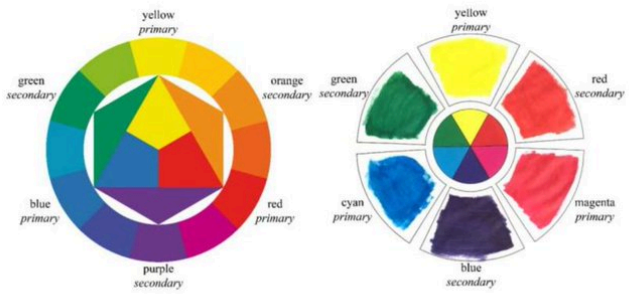
During primary school, we were all taught that there were three primary colours (📰):
🟥 Red
🟨 Yellow
🟦 Blue

We were then taught the following with amazement:
🟦+🟨=🟩
🟦+🟥=🟪
🟨+🟥=🟧
🟨+🟥+🟦=🟫

Did you know that this was a lie?
The actual subtractive primary colors were: Show more

Did your primary school teacher lie to you about color?

by Neil Dodgson, Victoria University of Wellington



Credit: Victoria University of Wellington

At primary school you were taught that there are three primary colors: red, yellow and blue, and that you can mix these to make all other colors. This is not true. Or rather, it is only a rough approximation to the truth. My recent article, published in the *Journal of Perceptual Imaging*, digs into the history of color wheels and color mixing to find that the truth is more complex and more interesting.

5:03 PM · Apr 20, 2025

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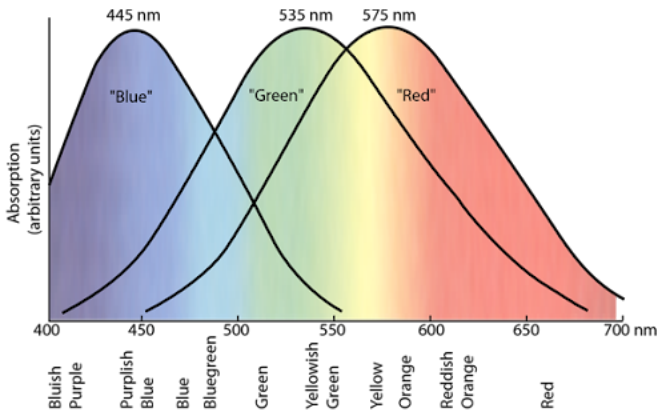

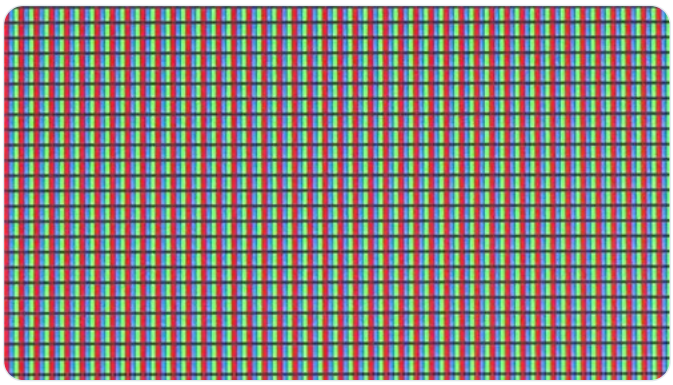

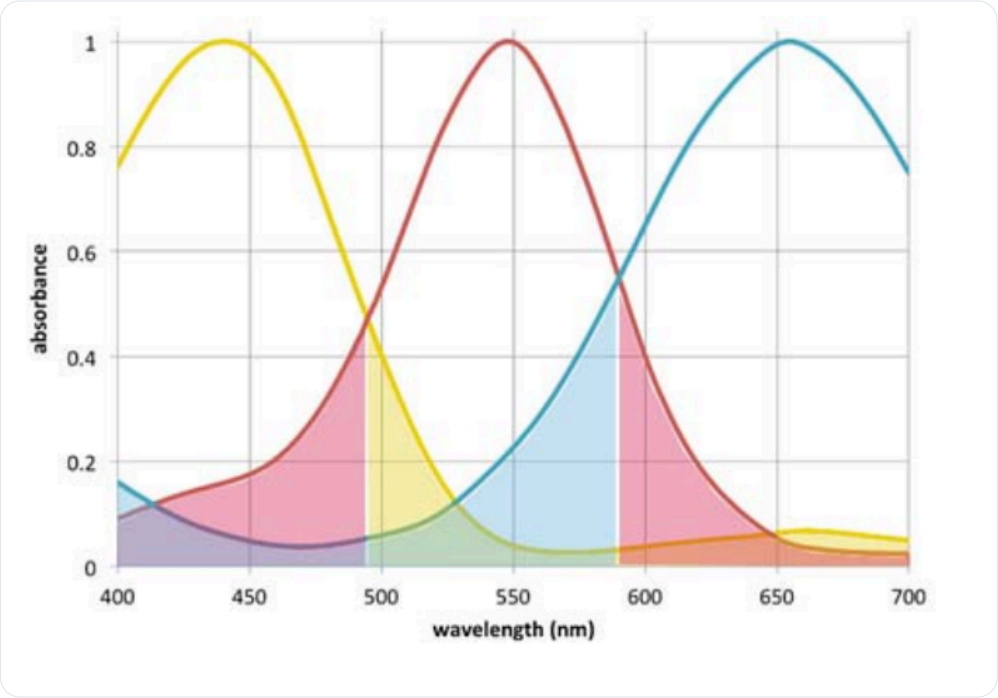
1
0.8
0.6
0.4
0.2
0
absorbance
400
450
500
550
600
650
700
wavelength (nm)

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